

WELLSENSE · DESIGN DECISION RECORD

WellSense TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WLS-DDR-001	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/decisions/0001-trl2-review-decisions.md @ 81641d4		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items D1 to D8 are adopted for TRL 3 work pending Amish's review; items O1 and O2 remain "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis WLS-PRC-001 v0.2 listed five key design choices, all marked proposed. On 2026-09-25 Amish asked for this batch of Design Molecule repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carried a recommendation is therefore adopted as recommended for TRL 3 work, open for his review. Items without a recommendation stay open. Nothing here is recorded as decided or approved by Amish.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in WLS-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Adopted recommendation	Status
D1	Budget treatment of FieldNode (review item 1)	Option (a): the FieldNode core is costed in the FieldNode repo, and the \$180 <code>budget_usd</code> covers the WellSense parts only; the complete logger cost is stated alongside. <code>budget_usd</code> is unchanged.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D2	Sensor type (review item 2; precis choice 1)	Option A: vented (gauge) 4 to 20 mA transducer for the first build; option B, a sealed absolute sensor with RS-485 and surface barometric compensation, kept as a variant	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Range (review item 3)	0 to 10 m of water; 0 to 20 m kept as a variant for wells with large pumping drawdown	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Access tube (review item 4; precis choice 2)	Own access tube wherever a pump or riser shares the casing; probe hung freely where there is no pump	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Default interval (review item 5)	15 min, with 1 min during pumping tests	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	Manual tape check (review item 6; precis choice 4)	Tape datum at installation and a tape check each quarter as part of the method, with one shared tape per community or partner	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	Data ownership (review item 7; precis choice 5)	Levels, time and well ID only; the hosting community controls sharing; CSV export	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D8	FieldNode as the core (precis choice 3)	Reuse the lab's shared FieldNode core for enclosure, power, radio and storage; WellSense designs only the well side and the interface	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

No rewording of the pitch or problem line was recommended at TRL 2, so both stay as they are in `project.yaml` and `README.md`.

Table 2. Items that remain open.

#	Item	Status
O1	First partner and region for co-design (an Indian farmer group, an African handpump programme or a US groundwater agency). No preference stated and no recommendation made.	Proposed, awaiting Amish
O2	Battery-only FieldNode variant without the panel (review item 9). Raised for discussion with the FieldNode project, with no recommendation to adopt; the baseline keeps the standard FieldNode core.	Proposed, awaiting Amish and the FieldNode project

■ Consequences

- `project.yaml`: only the TRL fields and the evidence list change. `budget_usd` stays at \$180, and the pitch and problem are unchanged.

- WLS-PRB-001, WLS-PRC-001 and WLS-REQ-001 are revised to v0.3. The design choices in the precis are no longer described as "proposed"; they are adopted for TRL 3 work pending Amish's review.
- Requirement R12 is restated under D1 as the WellSense parts per well at the 30 m design case, with the FieldNode core costed in the FieldNode repo (FND-CAL-001, \$126.00). R1 names the 0 to 10 m range (D3) with the 0 to 20 m variant, and R6 states that 1 min readings are logged and sent in batches (D5). No other target changes.
- The TRL 3 calculations (WLS-CAL-001) made detail choices within D2 that Amish should see: the FieldNode 12 V rail cannot drive the current loop through a 150 Ω shunt to the transducer's 12 V minimum, so the interface board gains a 12 to 24 V boost and a 3.3 V regulator; the transducer is specified in the 0.25 % class, since the 0.5 % class cannot reach R4 even after calibration; and calibration is two-point, by lifting the probe a measured 1 m in its tube. Within D8, the post is lengthened so the FieldNode core sits at its 1.75 m mounting height (FND-DDR-001, D11), and a footing concrete line is added to the BOM.
- The calculations also raise new items that need Amish's decision (cost over \$180 at 30 m, access tube bore, surface cable protection, hot-site temperature rating, pumping-test airtime). They are listed in `docs/REVIEW.md` as proposed and are not decided by this record.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.